

FINAL YEAR RESEARCH PROJECT

End-term Presentation

A Modular Framework for Prompt Injection Detection in GPT-Based AI Systems



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Presentation Outline

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 - Problem Statement
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- Literature Review
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Problem Statement

- Large Language Models (LLMs) are widely used in chatbots, coding assistants, and AI applications.
- Prompt Injection Attacks manipulate prompts to bypass AI safety mechanisms.
- These attacks can lead to:
 - Data Leakage
 - Unsafe Outputs
 - Unauthorized Actions
 - System Manipulation
- Existing defenses struggle against semantic and obfuscated attacks.
- A robust detection framework is required before prompts reach the LLM.

Problem Statement

Existing Challenges

- Prompt injection attacks bypass LLM safety instructions.
- Traditional filtering methods fail against advanced attacks.
- Transformer-only models are computationally expensive.
- Real-time AI systems require fast and reliable protection.

Need

A lightweight and intelligent framework capable of:

- Detecting malicious prompts early
- Reducing false negatives
- Improving semantic understanding
- Supporting real-time deployment

Motivations

- AI is increasingly used in everyday systems.
- Ensuring safety and reliability is critical.
- Attackers are constantly evolving to bypass protections.
- There is a need for a simple and effective solution.
- The system should:
 - Detect suspicious prompts early
 - Assign a risk score
- Goal:
 - Improve security
 - Build trust
 - Ensure safe usage for users and organizations

Literature Review

Table 1: Summary of Existing Approaches and Research Limitations

S. No.	Author & Year	Methods/Techniques Used	X`	Research Outcomes	Findings
1.	Narek Maloyan et al. (2026)	3D taxonomy (delivery vector, attack modality, propagation), system-level analysis	Agentic AI tool-integrated datasets	Identified vulnerabilities in agent-based systems	LLMs cannot distinguish instruction vs data; need pre-execution validation
2.	McHugh et al. (2025)	LLM classifier, rule-based filtering, sandboxing (layered defense)	Hybrid attack datasets (XSS + prompt injection)	Evaluated hybrid threat scenarios	Traditional defenses fail; layered approach improves detection
3.	Yulin Chen et al. (2025)	ML classifier, vector embeddings, multi-signal detection	Benchmark datasets	Improved robustness under adaptive attacks	Single-model defenses fail; combined signals work better
4.	Lan et al. (2025)	Keyword filtering, similarity search, BERT-based classification	Real-time LLM application datasets	Built real-time detection system	Hybrid detection improves usability and performance
5.	Shaheer et al. (2025)	Dataset curation (HackAPrompt), ML classifiers	HackAPrompt dataset	Improved detection coverage and robustness	Diverse datasets improve generalization across attacks





S. No.	Author & Year	Methods/Techniques Used	Dataset Used	Research Outcomes	Findings
6.	Xiaogeng Liu et al. (2024)	Momentum-based Greedy Coordinate Gradient (M-GCG) optimization	Synthetic + real prompt datasets	Generated universal and scalable attacks	Attacks generalize across tasks; current defenses lack robustness
7.	Robey et al. (2024)	Randomized smoothing, prompt perturbation (swap, insert, patch), response aggregation	JailbreakBench datasets	Reduced attack success rate significantly	Small input perturbations help detect adversarial prompts but impact performance
8.	Ayub & Majumdar (2024)	TF-IDF, Random Forest, Logistic Regression, XGBoost, OpenAI & MiniLM embeddings	467K+ prompt dataset	Compared ML classifiers for detection	Random Forest + embeddings outperforms deep models; strong pre-LLM screening
9.	Rahman et al. (2024)	Multilingual BERT, DistilBERT embeddings + SVM, RF, LR, NB	Custom multilingual dataset	Achieved 96.55% accuracy	Transformer embeddings + ML improves detection significantly
10.	Li et al. (2024)	DistilBERT, DeBERTa fine-tuning, MOF (Mitigating Over-defense)	NotInject benchmark	Improved detection accuracy by 30.8%	Reduces false positives while maintaining strong detection
11.	Donghyun Lee et al. (2024)	LLM tagging, inter-agent communication tracking	Multi-agent system datasets	Studied propagation of attacks	Prompt injection spreads across agents like a virus



S. No.	Author & Year	Methods/Techniques Used	Dataset Used	Research Outcomes	Findings
12.	Chen et al. (2024)	Instruction fine-tuning, delimiter-based structured query parsing	Prompt injection benchmarks	Reduced injection success significantly	Separating instructions from user data improves security
13.	Jiang Han (2024)	Adversarial prompt evaluation (political, offensive, misleading cases)	ChatGPT test scenarios	Evaluated real-world safety risks	Carefully crafted prompts bypass safety controls
14.	Kokkula et al. (2024)	Multi-layer framework (rule-based + BERT + companion LLM)	Custom datasets	Improved accuracy and reduced false negatives	Combined system performs better than individual models
15.	Yupei Liu et al. (2023)	Formal attack framework, Combined Attack strategy, benchmarking using ASV & MR metrics	Multi-LLM benchmark datasets	Compared multiple attack/defense strategies	Advanced prompt injection can override model behavior; existing defenses fail to detect complex attacks
16.	Jain et al. (2023)	Perplexity filtering, paraphrasing, retokenization, anomaly detection	Adversarial attack datasets	Combined multi-signal detection	Best results from combining classifier score + perplexity + length

Research Gap & Improvements

Research Gap

- Most existing systems rely on single-model detection techniques.
- Traditional methods fail against semantic and obfuscated prompt injection attacks.
- Transformer-only models increase computational cost and latency.
- Many approaches produce high false positives and false negatives.
- Existing systems mainly provide binary safe/unsafe classification.
- Limited support for real-time deployment in GPT-based applications.

Proposed Improvements

- Hybrid Ensemble + Transformer Framework
- TF-IDF with LR, SVM, and SGD for fast detection
- XLM-RoBERTa for semantic and contextual understanding
- Reduced false negatives in complex attacks
- Risk-based ALLOW / REVIEW / BLOCK classification
- Lightweight and deployable real-time AI security framework

Proposed Solution & Architecture



Modular Framework for Prompt Injection Detection in GPT-Based AI Systems

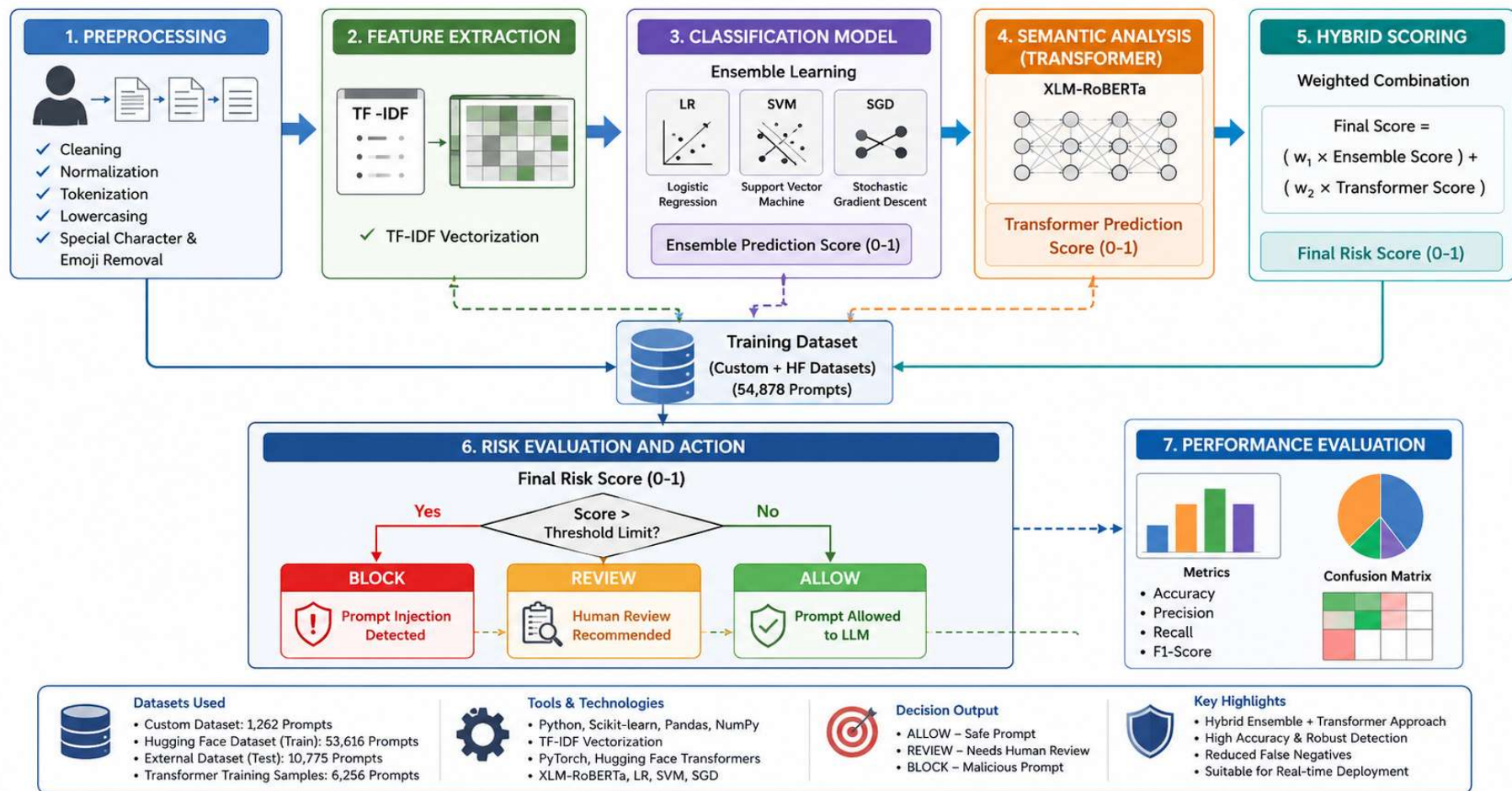


Fig 1 : Schematic Model Diagram of the Proposed Model

Dataset & Tools/Technologies

Dataset

- Created a hybrid dataset using custom and Hugging Face prompt injection datasets.
- Developed 1,262 manually collected prompts with benign and malicious samples.
- Used 53,616 prompts for ensemble model training and 10,775 prompts for external testing.
- Preprocessed and balanced the data, selecting 6,256 samples for XLM-RoBERTa training.

Tools & Technologies

- Python with Scikit-learn, NumPy, and Pandas for data processing and machine learning.
- PyTorch and Hugging Face Transformers for deep learning and XLM-RoBERTa implementation.
- TF-IDF Vectorization with Logistic Regression, SVM, and SGD for ensemble learning.
- Jupyter Notebook / Google Colab with evaluation metrics like Accuracy, Precision, Recall, F1-Score, and Confusion Matrix.

Experimentation

- Applied preprocessing techniques such as lowercasing, emoji removal, and special character cleaning.
- Converted prompts into numerical vectors using TF-IDF vectorization.
- Trained ensemble classifiers: Logistic Regression, SVM, and SGD.
- Fine-tuned XLM-RoBERTa for semantic prompt injection detection.
- Combined ensemble and transformer outputs using weighted scoring.
- Implemented threshold-based decisions: ALLOW, REVIEW, and BLOCK.

Experimentation

Ensemble Learning Model

- Accuracy: 97.21%
- Precision: 97.95%
- Recall: 96.89%
- F1-Score: 97.42%

Hybrid Framework (Ensemble + XLM-RoBERTa)

- Accuracy: 96.93%
- Precision: 96.69%
- Recall: 97.70%
- F1-Score: 97.19%
- Reduced false negatives and improved detection of obfuscated attacks.
- Better semantic understanding and robustness against context-aware prompt injections.

Conclusion

- Proposed a hybrid prompt injection detection framework combining ensemble learning and transformer-based semantic analysis.
- TF-IDF and ensemble models provided fast and reliable prompt classification.
- XLM-RoBERTa improved semantic understanding and detection of obfuscated attacks.
- Reduced false negatives and improved robustness against malicious prompts.
- Achieved a balance between detection accuracy, security, and practical deployment capability.



Future Scope

- Expand the dataset with more diverse and adversarial prompt injection samples.
- Perform advanced robustness and real-time testing techniques.
- Explore lightweight architectures for faster real-time deployment.
- Develop low-latency APIs or plugins for production AI systems.
- Add explainability features for better decision interpretation.
- Extend the framework for multilingual prompt injection detection.
- Implement continual learning to adapt against evolving attacks.



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*Thank
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